Vikramaditya Roy

Machine Learning & Backend Systems Engineer

Roll No: 22AI015 | Department: AI & Data Science | CGPA: 9.30 / 10.0

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PROFESSIONAL SUMMARY

AI researcher and engineer with deep knowledge in deep neural networks, transformer models, FastAPI microservices, and high-performance vector databases.

CORE TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, Bash Scripting
- **ML & Deep Learning:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, OpenCV
- **Backend & MLOps:** FastAPI, Flask, Docker, MLflow, Triton Server, FAISS, PostgreSQL
- **Core Competencies:** Vector Embeddings, LLM Fine-Tuning, Computer Vision, Model Quantization

WORK EXPERIENCE & INTERNSHIPS

AI & ML Engineering Intern | *DeepCognition Labs* May 2024 - Aug 2024

- Fine-tuned transformer models for multi-label technical document classification achieving 94.2% F1 score.
- Deployed containerized inference pipelines on AWS EC2 GPU instances with sub-50ms latency.

FEATURED TECHNICAL PROJECTS

Multimodal Semantic Search Engine *(Python, FastAPI, PyTorch, FAISS, Docker)*

- Indexed 500,000+ academic papers with hybrid keyword and dense vector similarity search.
- Implemented re-ranking pipeline improving retrieval precision by 28% over pure BM25 search.

Real-Time Industrial Defect Detection *(Python, PyTorch, OpenCV, Flask)*

- Trained YOLOv8 object detector achieving 97.4% mAP for assembly line automated quality control.

EDUCATION & ACADEMIC BACKGROUND

Bachelor of Technology in AI & Data Science • University School of Information Technology 2022 - 2026

CGPA: 9.30 / 10.0 | Relevant Coursework: Machine Learning, Deep Neural Networks, Applied Linear Algebra, Probability

CERTIFICATIONS & HONORS

- DeepLearning.AI Deep Learning Specialization (Andrew Ng)
- NVIDIA Deep Learning Institute Certificate: Fundamentals of Deep Learning